Biology-1: Lecture 06

(BIO103)

Chapter 2. Chemistry of life

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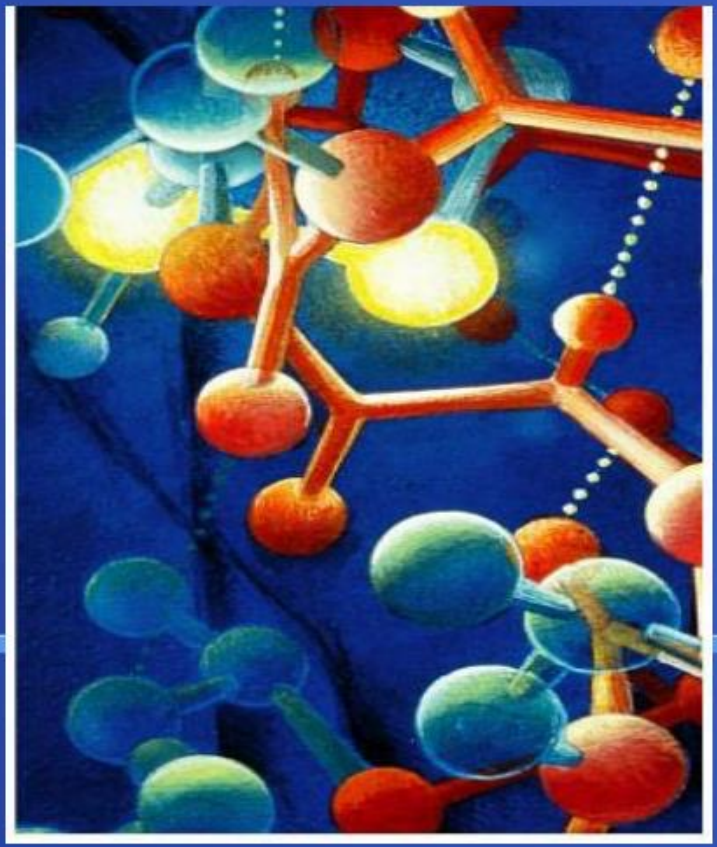
Chap 2. Chemistry of life

Lecture 6: Outlines Spring 2026

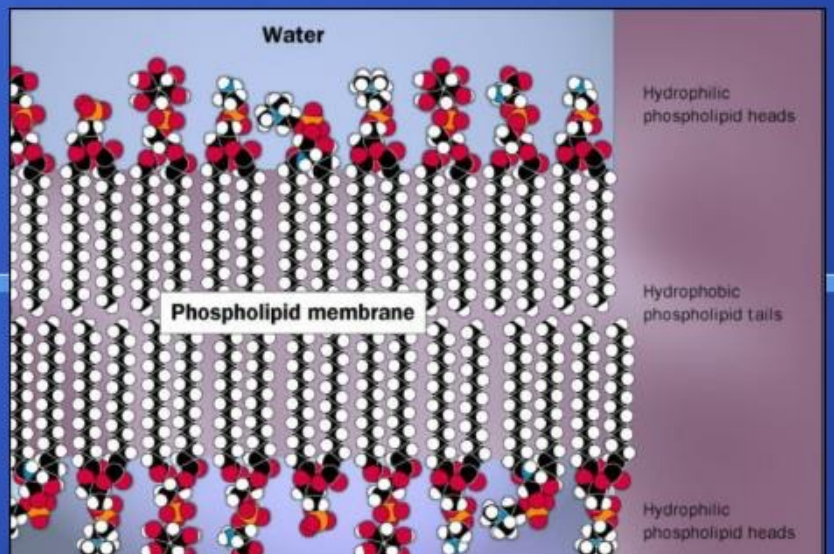
1. Basic Chemistry revisiting:

- Elements, atoms, molecules, compounds
- Structure of atoms,
- Chemical bonding
- Oxidation-reductions,
- Isotope and Radioisotope

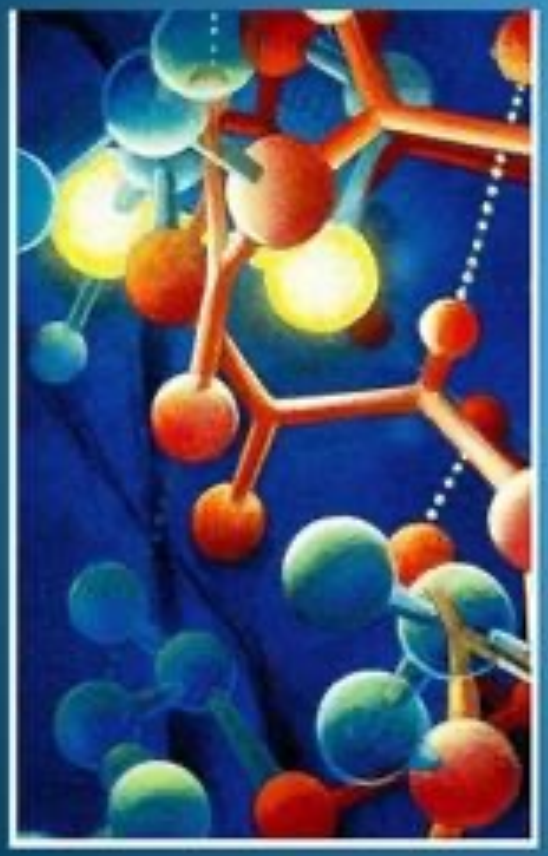
Ref: 1. Lehninger's Principles of Biochemistry, Chapter 1
2. Biochemistry of Dummies- John T Moore



The Chemistry of Life



Chemistry of Life



What are the building blocks of organisms?

Why do we have to eat?

Chemistry of Life

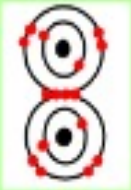
Life

Atom

Oxygen (O)

Molecule

Dioxide (O_2)



Macromolecule

Phospholipid

Cell

Clara Cell

Tissue

Epithelial tissue

Organ

Lung

Organ System

Respiratory System

Organism

Panthera leo (Lion)

Population

Pride of Lions

Community

Lions and Zebras

Ecosystem

Lions, Zebras, and the Environment

Biosphere

Life on Earth

Chemistry of life: Elements, Atoms, Molecules, compounds

- Everything – whether it is a rock, frog, or flower – is made of substances called elements.

- An **element** is a substance that **cannot** be broken down into simpler chemical substances.

Chemistry of life: Elements, Atoms, Molecules, compounds

Natural elements in living things

- Of the naturally occurring elements on Earth, only about 25 are essential to living organisms.

- Carbon, hydrogen, oxygen, and nitrogen make up more than 96 percent of the mass of a human body

Chemistry of life: Elements, Atoms, Molecules, compounds

Trace elements

- Trace elements such as iron and copper, play a vital role in maintaining healthy cells in all organisms.

- Plants obtain trace elements by absorbing them through their roots; animals get them from the foods they eat.

Essential Bio-elements

- Essential bio-elements are chemical elements that are constituents of cellular materials and that are necessary for the growth, maintenance and reproduction of a living organism.

Two types

1. Major:

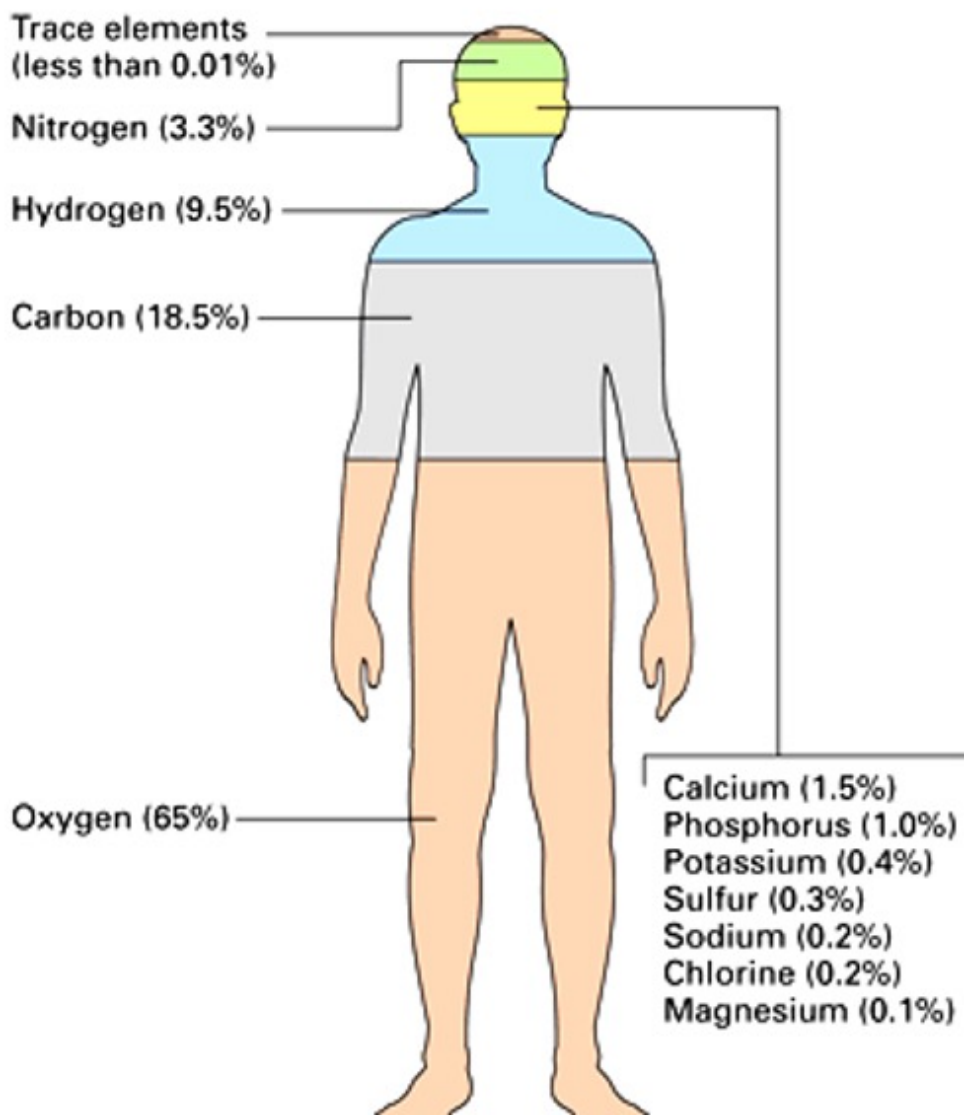
- Bio-elements which are needed in relatively higher concentration ($>10^{-4}$M)
- There are 12 such bio-elements

2. Minor

- Those which are needed in small amounts
- Critical to cell function

Elements of Life

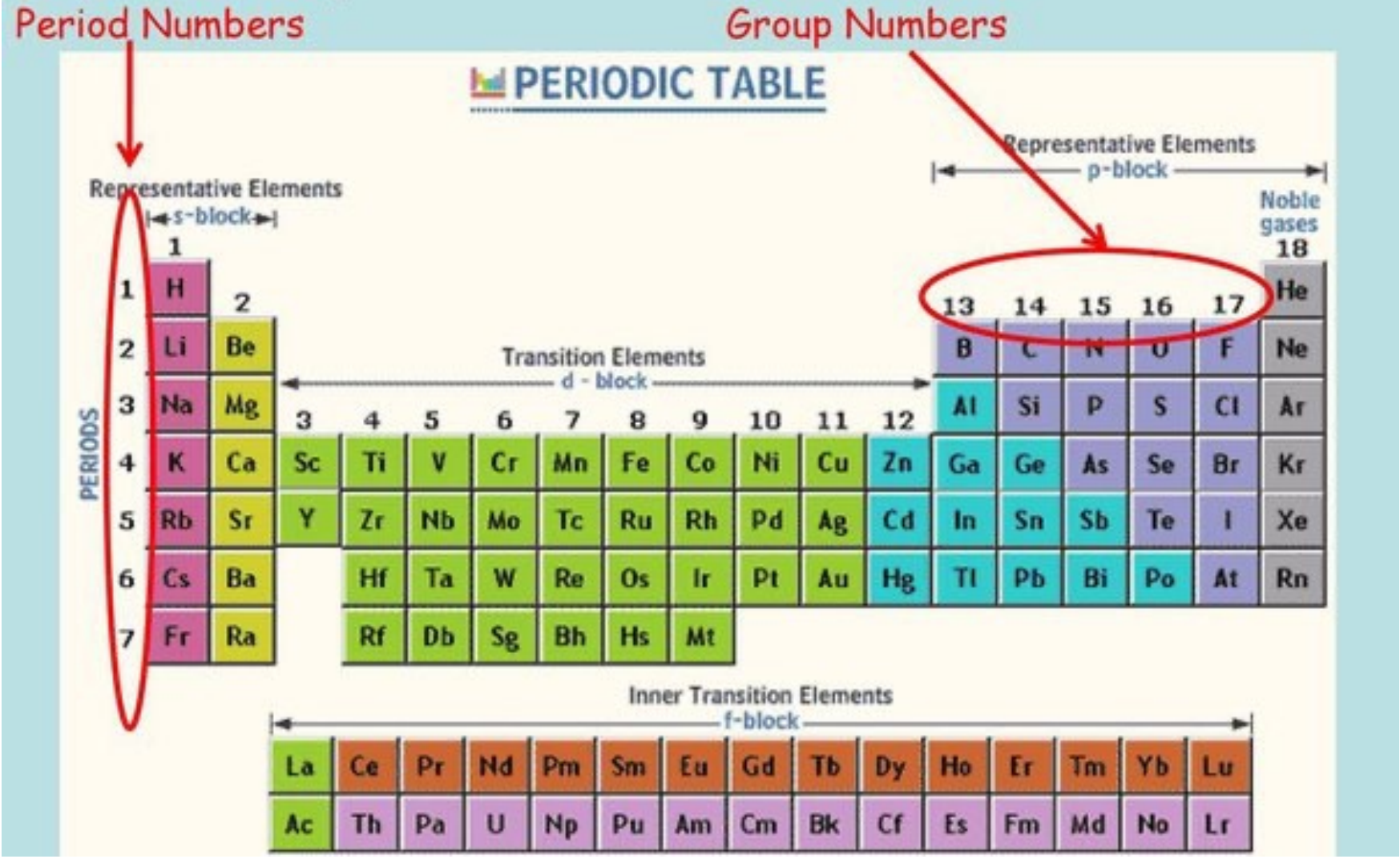
- 96% of living organisms is made of:
  - carbon (C)
  - hydrogen (H)
  - oxygen (O)
  - nitrogen (N)



Periodic Table of Elements

- Modern table devised by Demetri Mendeleev.
- Currently there are 118 named elements.
- Columns also called Families range from 1-18.
- The family number can also tell us the number of valence electrons, or the electrons in the outermost level.

Group and Period Numbers



Periodic Table of Elements

- Rows also called Periods range from 1-7.
- As you move across the table the elements become less and less reactive.
- The elements are arranged in order of increasing atomic number.
- They are also grouped by metals and nonmetals.

Metals, Nonmetals, and Metalloids

H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	
Fr	Ra	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Uub	-	Uuq	-	-	-	-

Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

metals

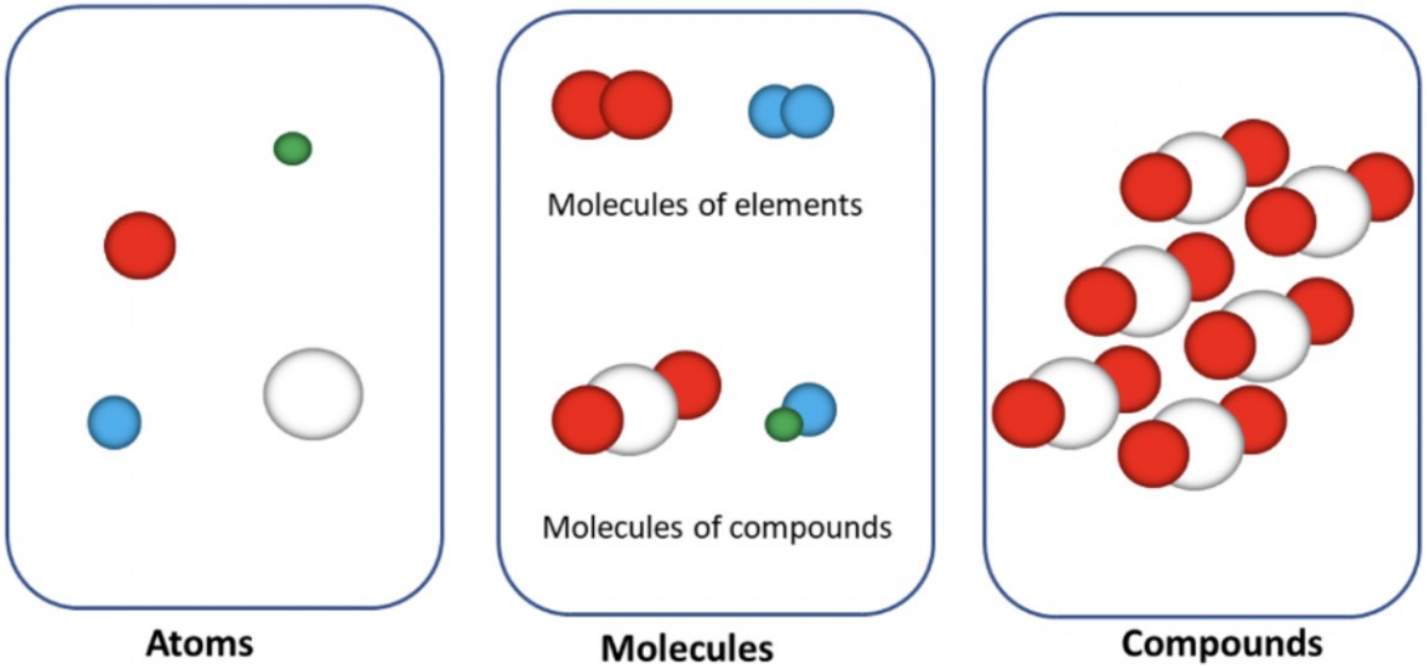
metalloids

nonmetals

Atoms: The Building Blocks of Elements

- An atom is the smallest particle of an element that has the characteristics of that element.

- Atoms are the basic building blocks of all matter.

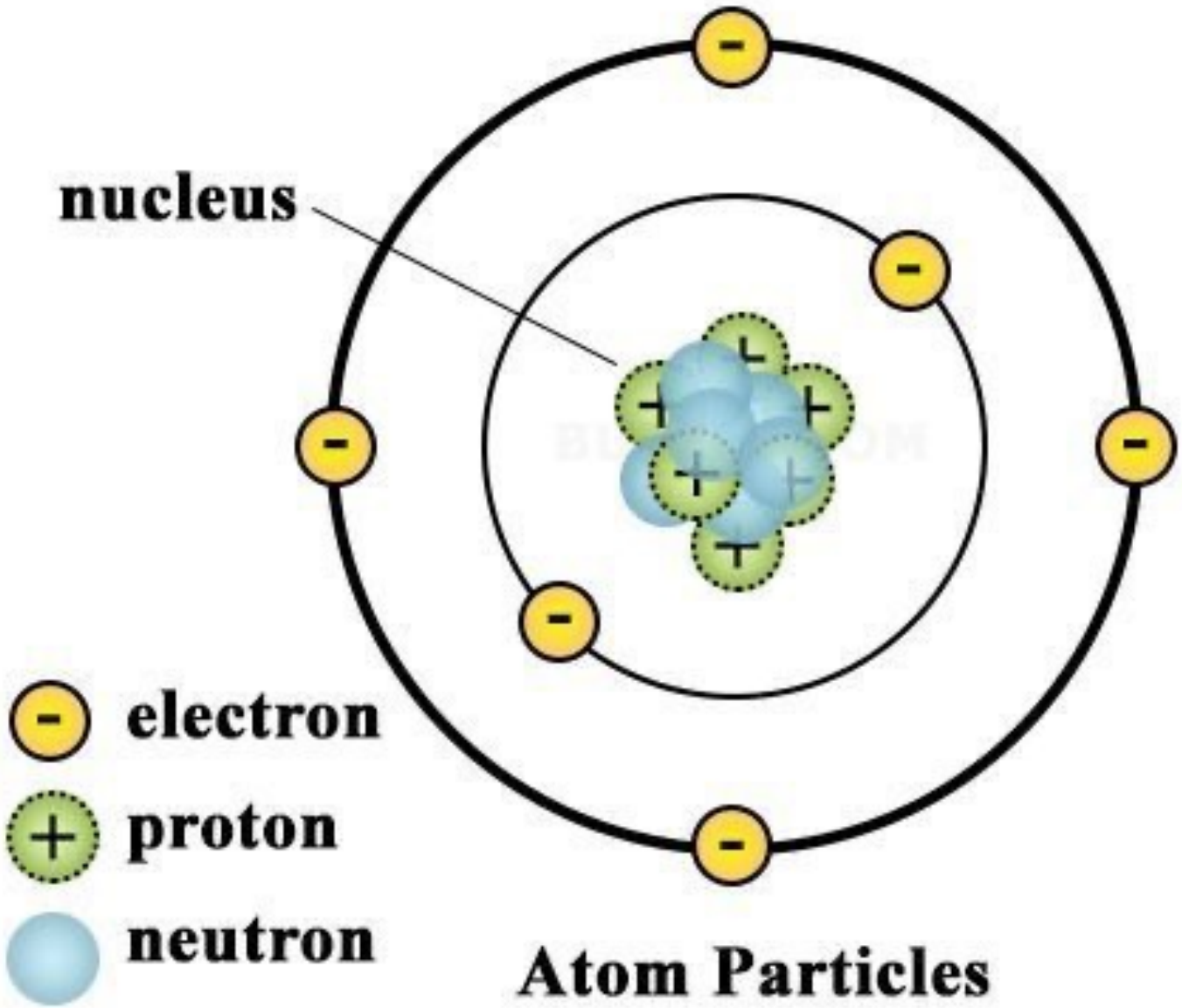


The Structure of an atom

- The center of an atom is called the nucleus.
- Nuclei contain positively charged particles called protons
- Most contain particles that have no charge, called neutrons.
- **Atomic mass** is equal to the protons + neutrons.
- The region of space surrounding the nucleus contains extremely small, negatively charged particles called electrons
- Because opposites attract, the negatively charged electrons are held in the electron cloud by the positively charged nucleus.

Chemistry of life : The Structure of an atom

- **Electron:** Negatively charged subatomic particle that occupies orbitals around the atomic nucleus.
- **Proton:** Positively charged subatomic particle that occurs in the nucleus of all atoms.
- **Neutron:** Uncharged subatomic particle in the atomic nucleus.
- **Nucleus:** Core of an atom; occupied by protons and neutrons.
- **Mass number:** Total number of protons and neutrons in the nucleus of an element's atoms.



Atom Particles

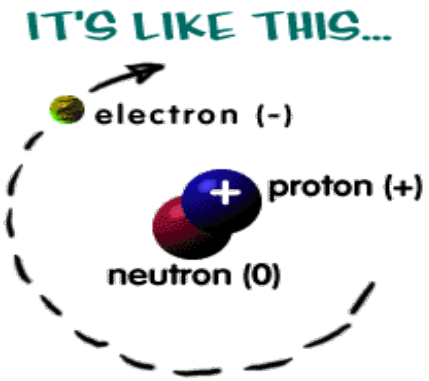
Chemistry of life : The Structure of an atom

- **What are the three basic parts of an atom?**

Electron = "-" negative charge

Proton = "+" positive charge

Neutron = neutral (a charge of zero)

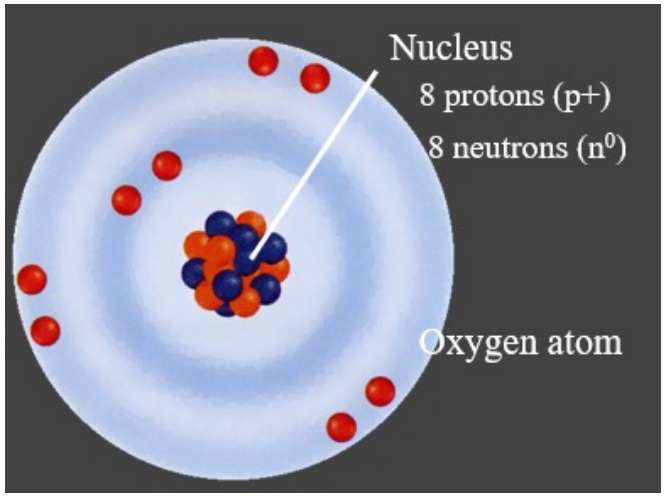


- Protons and neutrons always in the center of atom (the nucleus).

- Electrons are found whizzing around nucleus in areas called orbitals.

Q: **If there is an equal number of electrons and protons in an atom, what is it's charge?**

Electron Energy levels

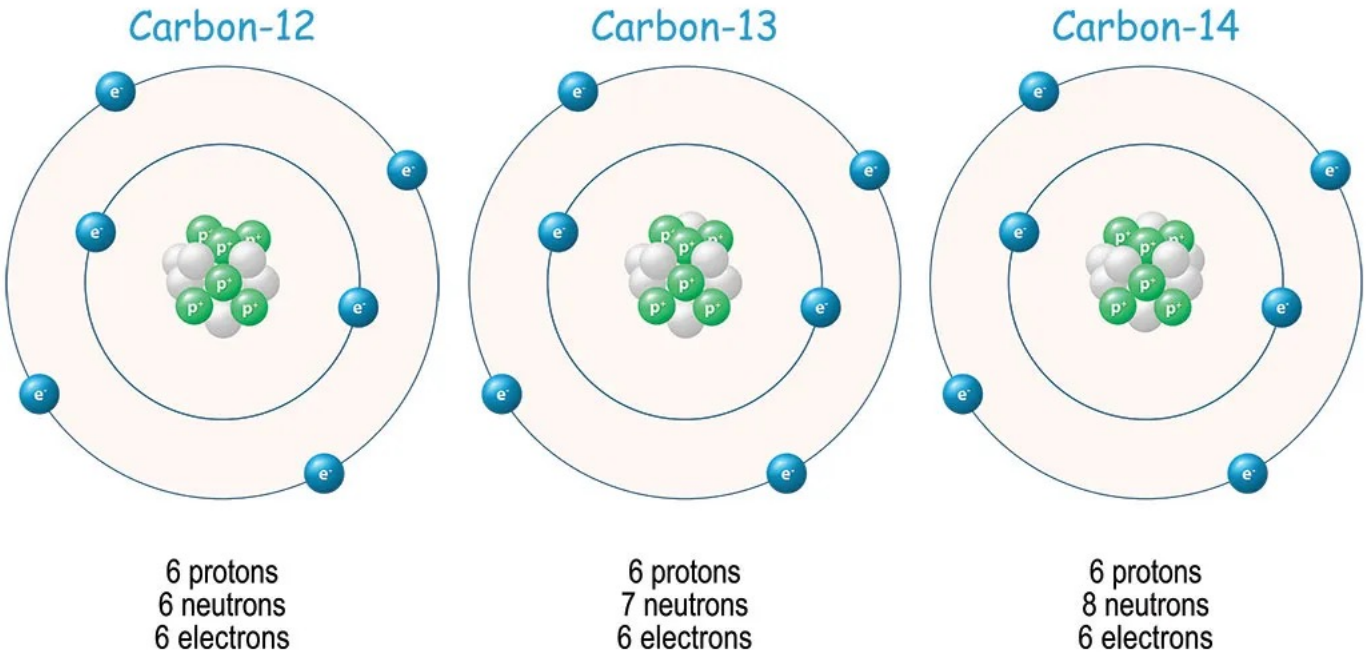


- Electrons exist around the nucleus in regions known as energy levels.
- The first energy level can hold only two electrons.
- The second level can hold a maximum of eight electrons.
- The third level can hold up to 18, 8 or less electrons.
- The fourth level can hold up to 32, 18, 8, or less electrons
- Atoms contain equal numbers of electrons and protons; therefore, they have no net charge or they are neutral.

The Structure of an atom: Isotopes

- Atoms of the same element always have the same number of protons but may contain different numbers of neutrons.
- Atoms of the same element that have different numbers of neutrons are called isotopes of that element.
- If the neutron number changes then the atomic mass also changes. The proton number stays the same.
- Examples: C-12, C-13, and C-14

Isotopes of Carbon

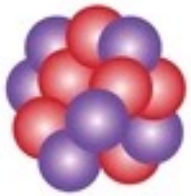


stable isotopes

unstable isotopes

Which one is radioactive isotope?

Isotopes of Carbon

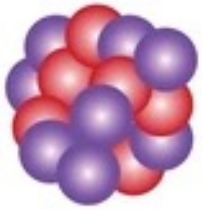


Carbon 12

6 Protons
6 Neutrons
6 Electrons

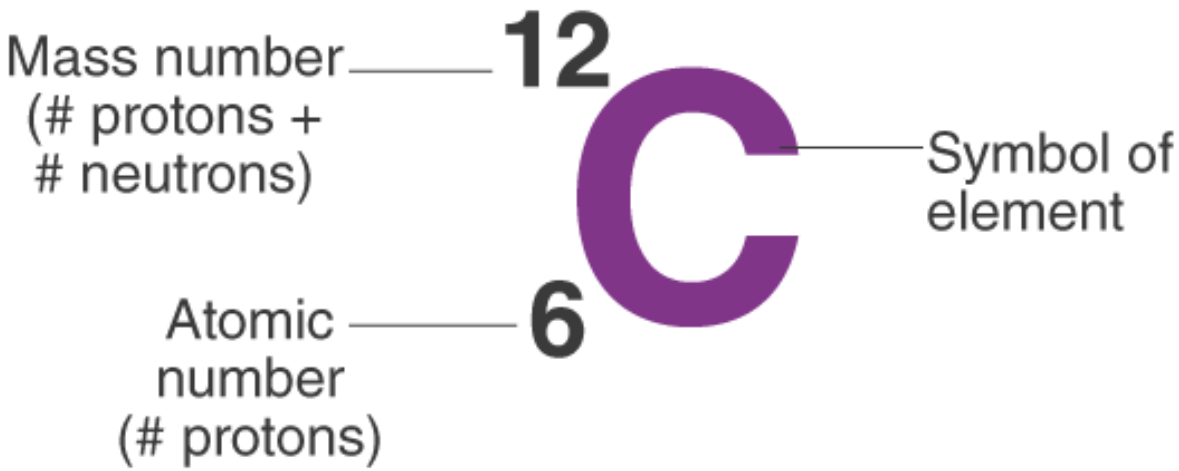
Carbon 13

6 Protons
7 Neutrons
6 Electrons

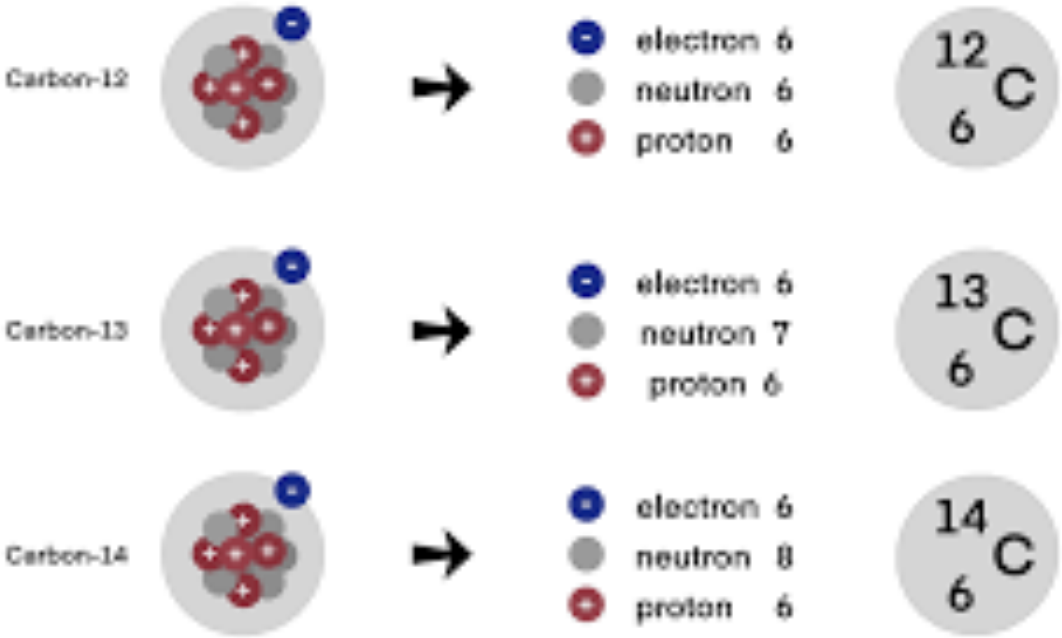


Carbon 14

6 Protons
8 Neutrons
6 Electrons


Mass number
(# protons +
neutrons)
12
C
Symbol of
element
Atomic
number
(# protons)
6

Isotope

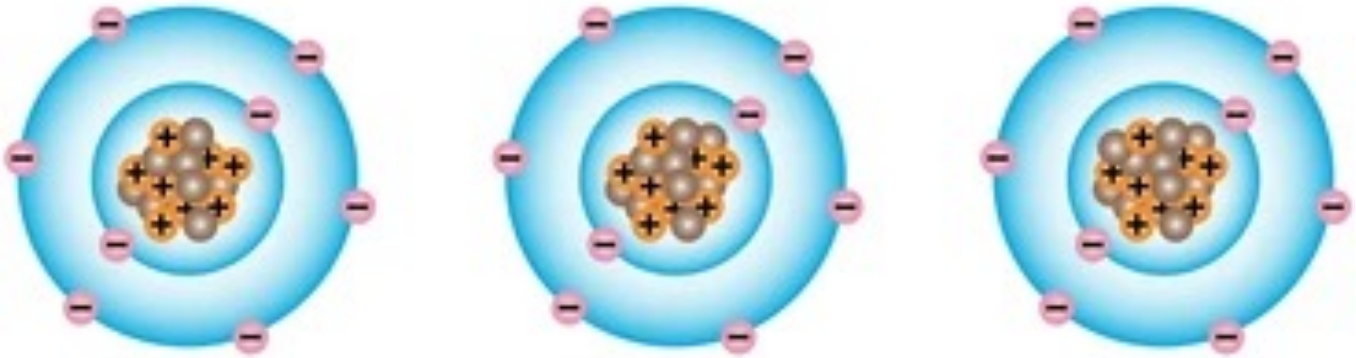

Carbon-12
electron 6
neutron 6
proton 6
12 C 6
Carbon-13
electron 6
neutron 7
proton 6
13 C 6
Carbon-14
electron 6
neutron 8
proton 6
14 C 6

Isotopes

- atoms have same number of protons but a different number of neutrons
  - same atomic number, different atomic weight
    - ex. oxygen forms isotopes (O^{16}, O^{17}, O^{18})
    - same element!

- *How many protons in O^{16}? O^{17}? O^{18}? How many neutrons?*
- unstable isotopes are radioactive and emit energy
  – used in medicine and research

Isotopes of Oxygen



Oxygen 16	Oxygen 17	Oxygen 18
8 Electrons	8 Electrons	8 Electrons
8 Protons	8 Protons	8 Protons
8 Neutrons	9 Neutrons	10 Neutrons
Nuclear Number = 8 + 8 = 16	Nuclear Number = 8 + 9 = 17	Nuclear Number = 8 + 10 = 18

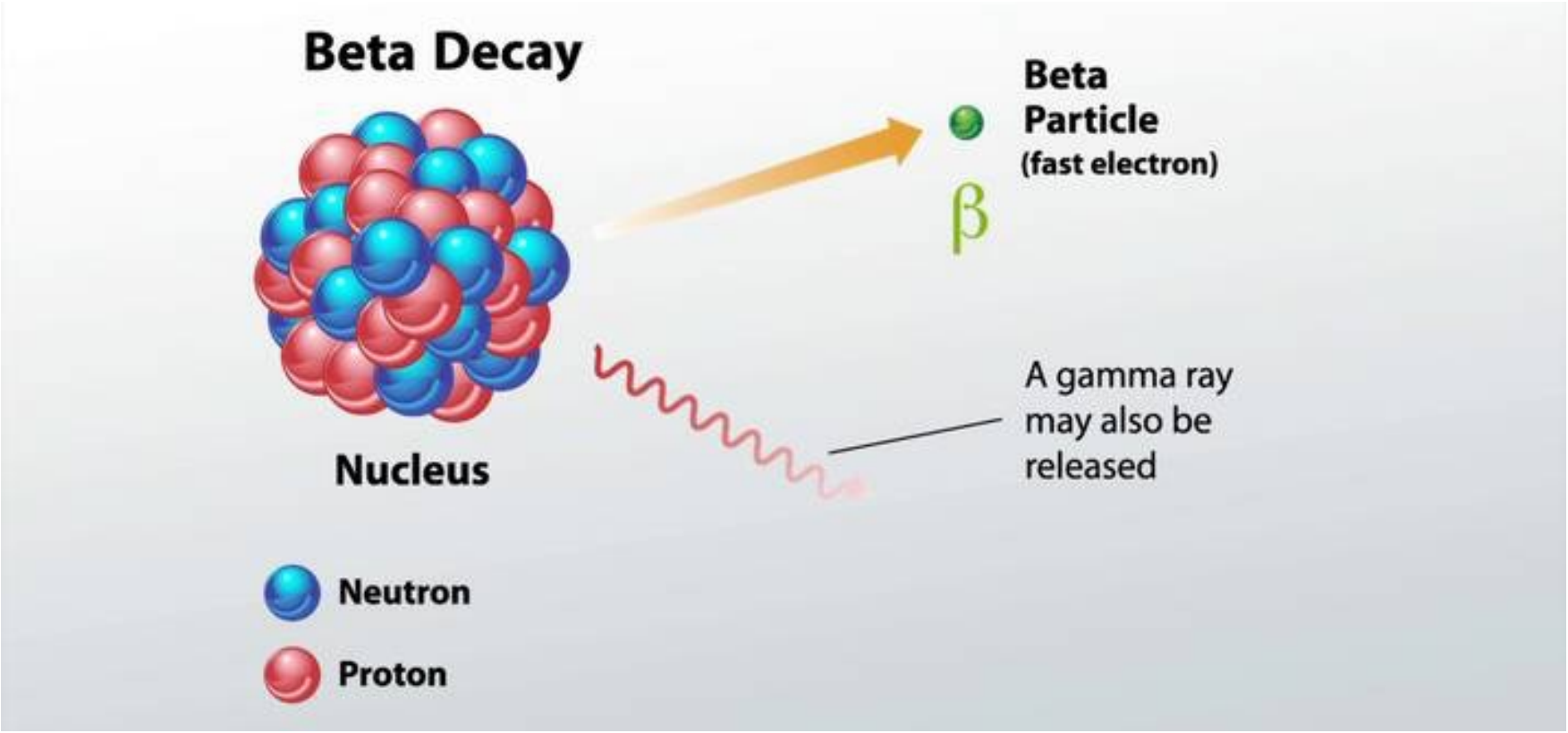
The Structure of an atom: Isotopes

- **Isotopes:** Forms of an element that differ in the number of neutrons their atoms carry.

- **Radioisotope:** Isotope with an unstable nucleus.

- **Radioactive decay:** Process by which atoms of a radioisotope spontaneously emit energy and subatomic particles when their nucleus disintegrates.

The Structure of an atom: Isotopes



Chemistry of life : The Structure of an atom

Isotopes/Radioisotopes

- Atoms of the most common carbon isotope, 12C, have six protons and six neutrons; those of 14C have six protons and eight neutrons (6 + 8=14).

- 14C (carbon 14) is a radioactive isotope, or radioisotope, which means that the nucleus of a 14C atom is unstable. Atoms of radioisotopes emit subatomic particles and energy when their nucleus spontaneously disintegrates. This process, radioactive decay, can transform one element into another.

- For example, 14C decays when one of its neutrons splits into a proton and an electron. The proton stays in the nucleus, and the electron is emitted. Thus, an atom of 14C (with six protons and eight neutrons) becomes an atom of 14N, which is nitrogen (with seven protons and seven neutrons).

Compounds and Bonding

- A compound is a substance that is composed of atoms of two or more different elements that are chemically combined.

- Atoms combine with other atoms only when the resulting compound is more stable than the individual atoms.

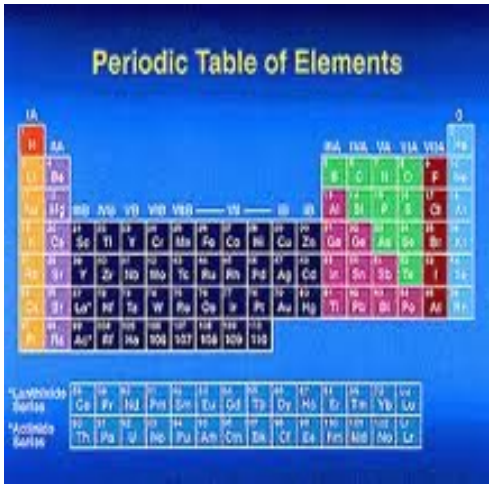
Compounds and Bonding

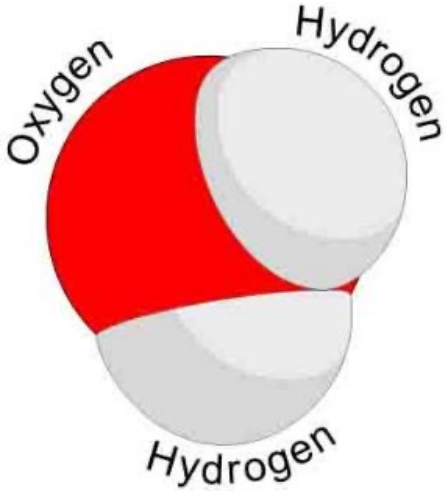
- Elements → A pure substance that consists only of atoms with the same number of protons.

- Atom → The smallest unit of an element.

- Molecule → Two or more atoms joined together chemically.

- Compound → Molecule containing at least two different elements.





Compounds and Bonding

Molecule –when two or more atoms chemically combine

Compound –when two or more atoms of different elements chemically combine in defined proportions.

Molecular formulas – elements present and the number of each atom present in the molecule

H_2 $C_6H_{12}O_6$ H_2O

Compounds and Bonding

- Examples of molecules: molecular hydrogen (H_2), molecular oxygen (O_2) and molecular nitrogen (N_2).

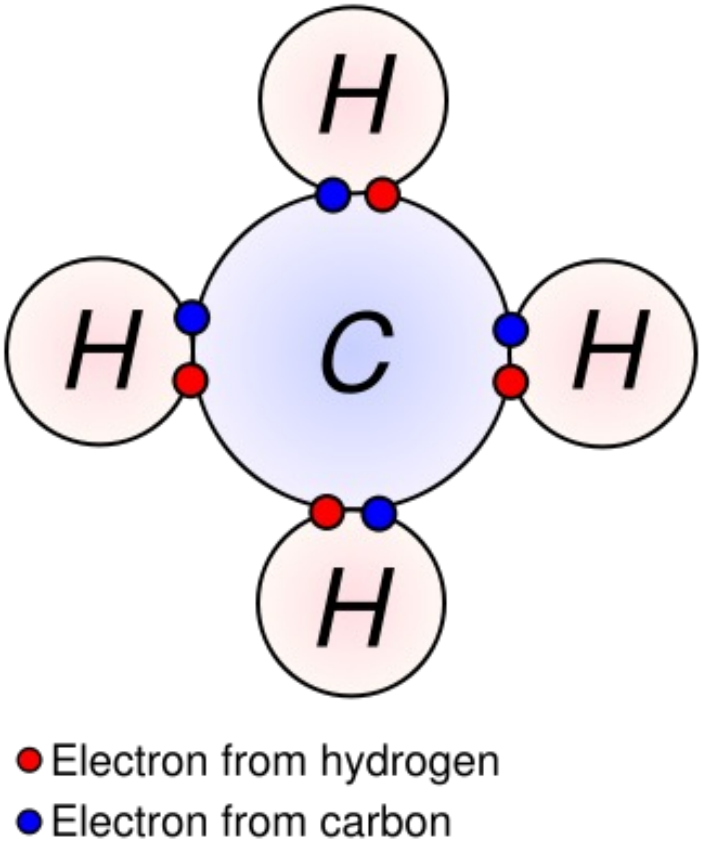
- Examples of compounds: Only molecules containing two or more elements, such as carbon dioxide (CO_2) and methane (CH_4).

Inorganic versus Organic molecules

○ Inorganic Molecules:
Molecules that don't have Carbon Hydrogen (C-H) bonds.

○The major organic macromolecules (big molecules with carbon-hydrogen bonds) found in living things are:

1. Carbohydrates
2. Proteins
3. Nucleic Acids
4. Lipids



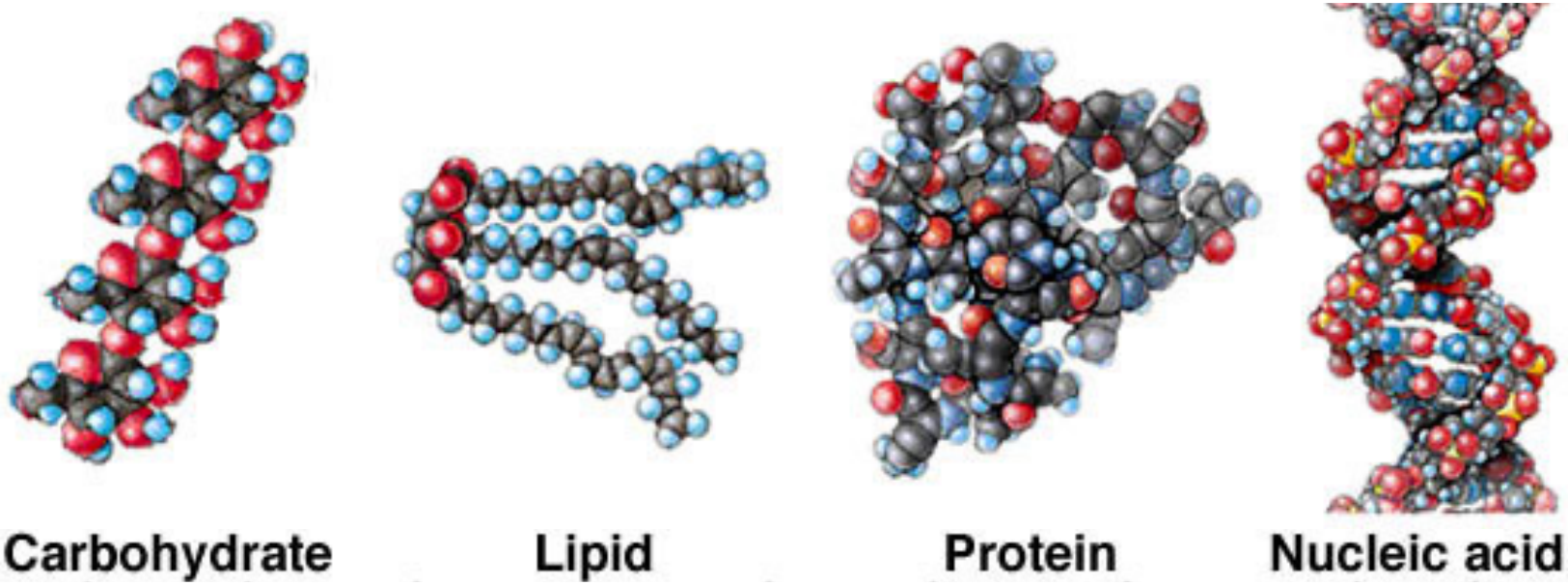
Molecules of life

- Put C, H, O, N together in different ways to build living organisms
- What are bodies made of?
  - carbohydrates
    - sugars & starches
  - proteins
  - fats (lipids)
  - nucleic acids
    - DNA, RNA

Molecules of life

How do you build a cell?

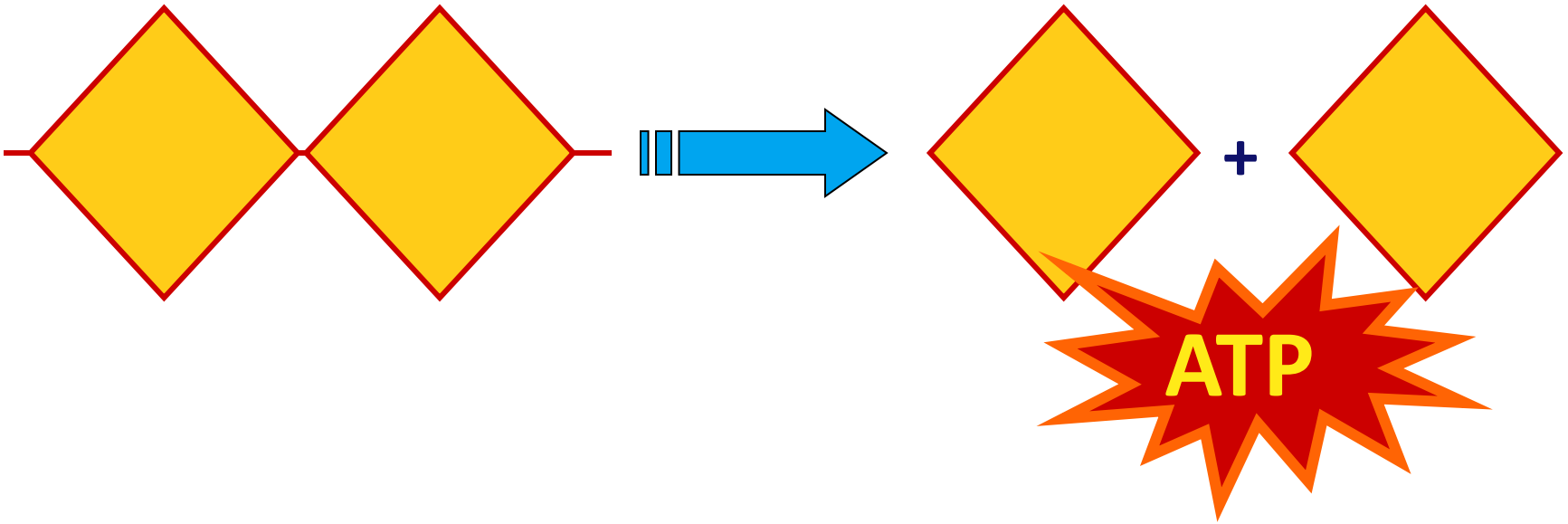
Start with water, add lots of small carbon-containing molecules and …….



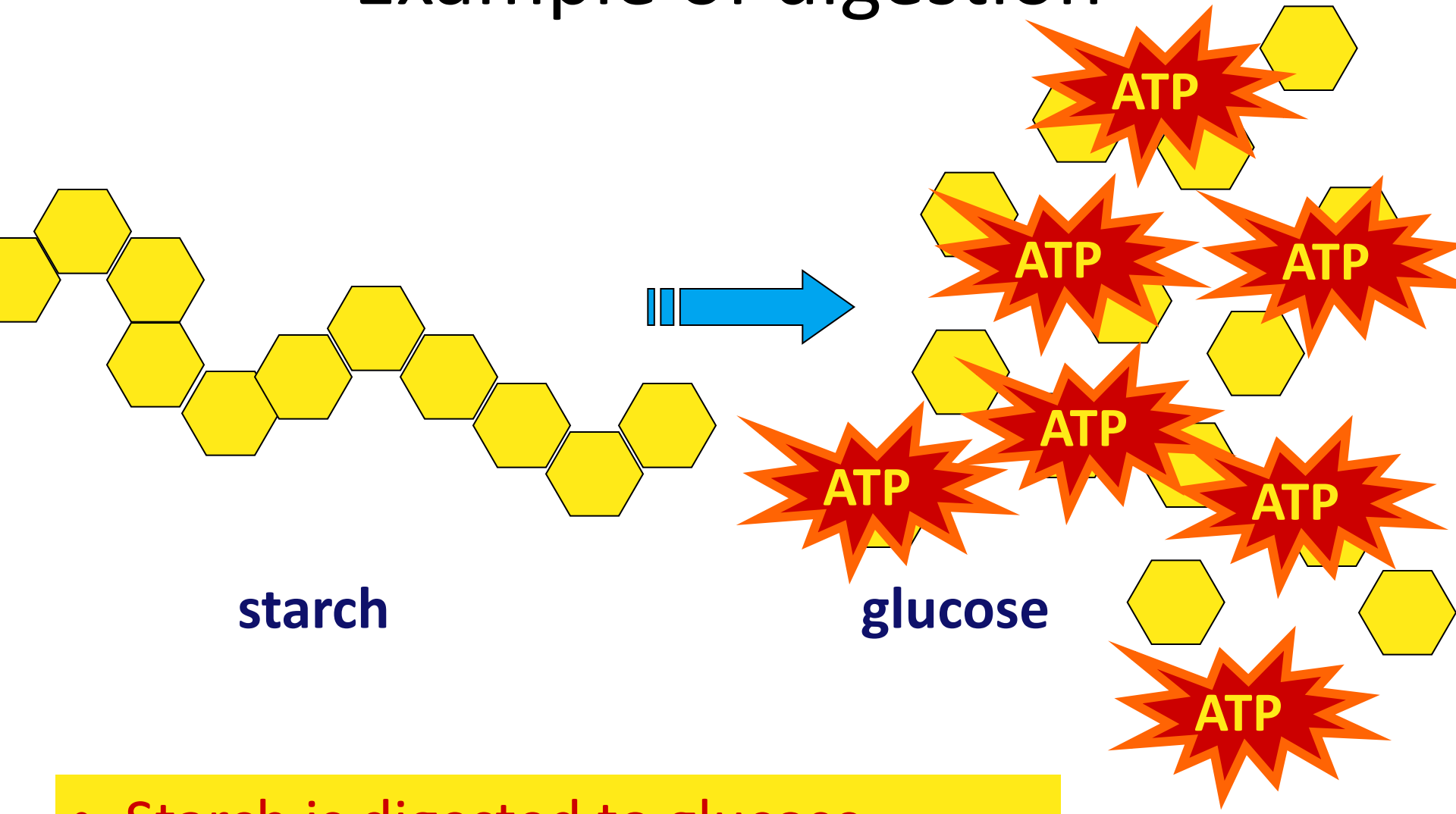
… these are the four major types of organic molecules.

How to take large molecules apart

- Digestion
  - taking big molecules apart
  - getting raw materials
  - making energy (ATP)

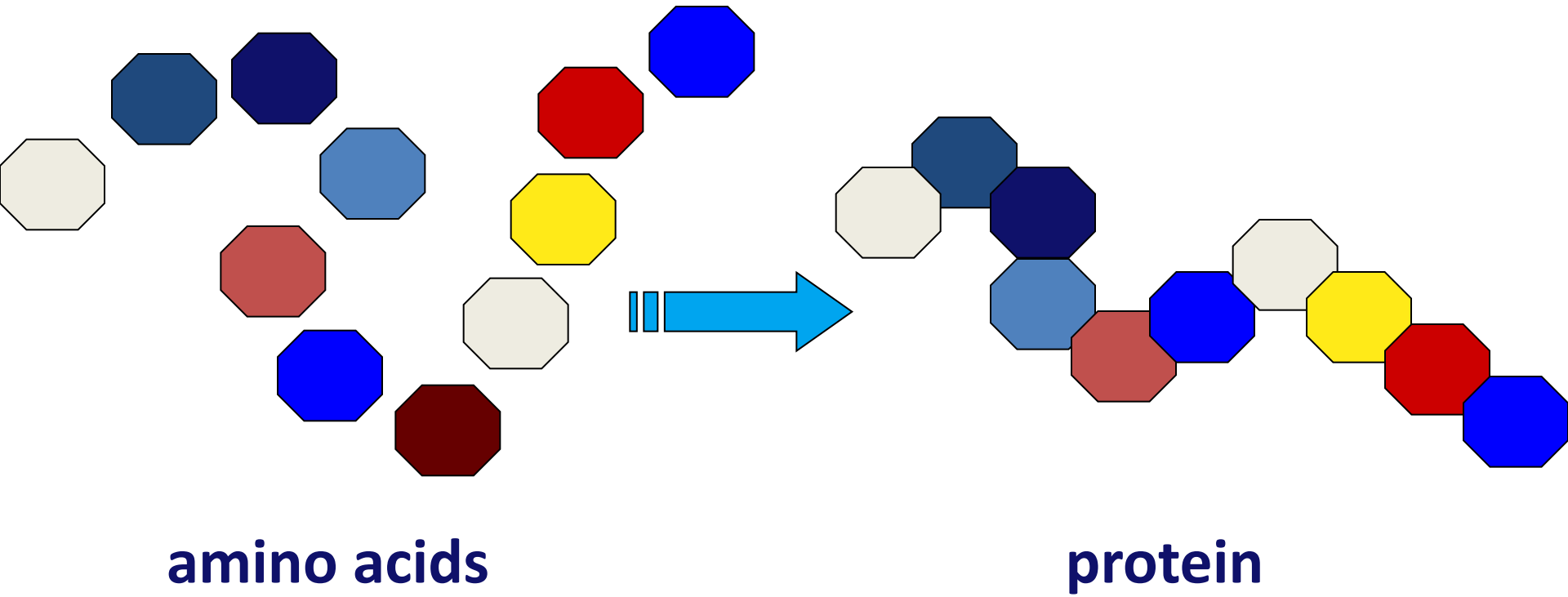


Example of digestion



- Starch is digested to glucose

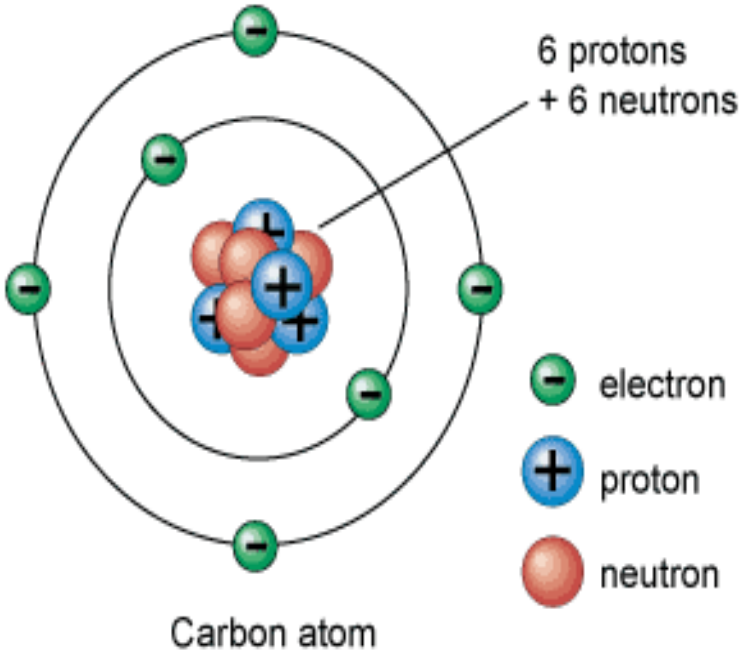
Example of synthesis

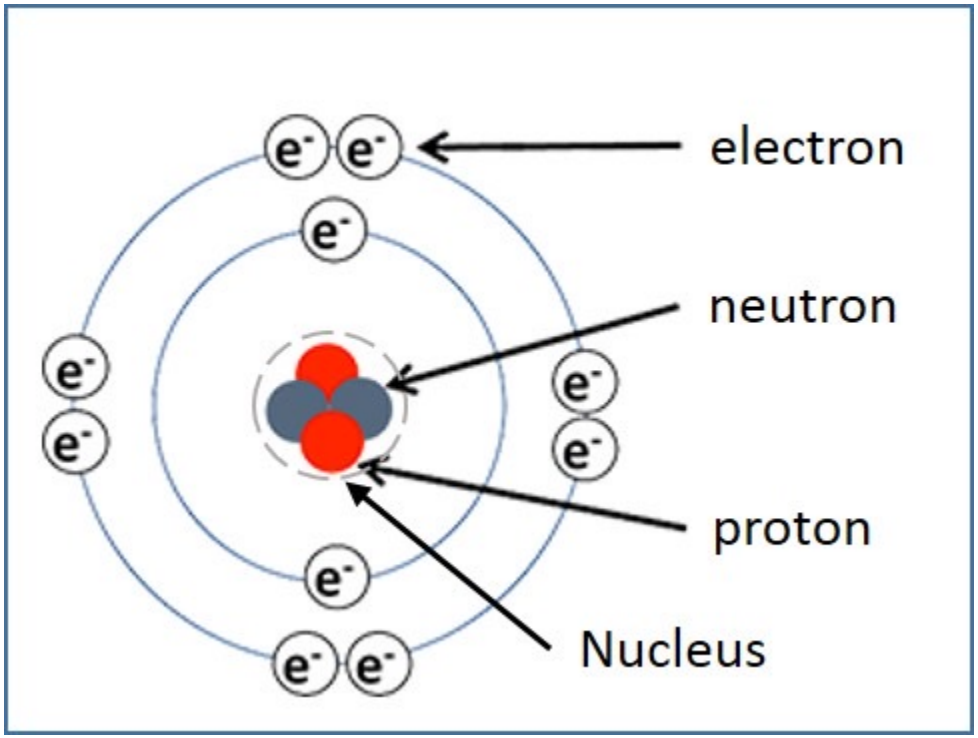


- Proteins are synthesized by bonding amino acids

Chemistry of life : Chemical bonding

- The electrons in an atom are located at different energy levels.
- Electrons in the highest energy level are called **valence electrons**.
- Number of valence electrons governs an atom's bonding behavior.
- Atoms are much more stable, or less reactive, with a full valence shell.
- By moving electrons, the two atoms become linked. This is known as chemical bonding.



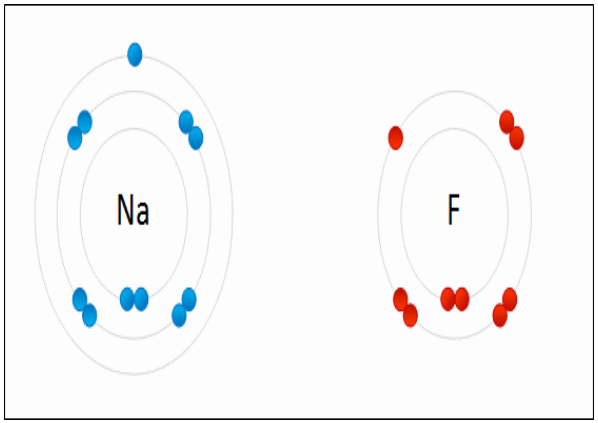


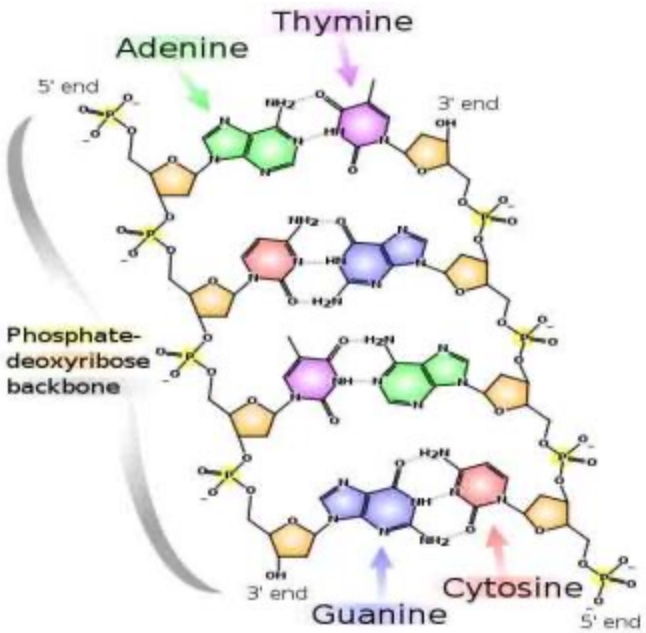
Structure of atoms

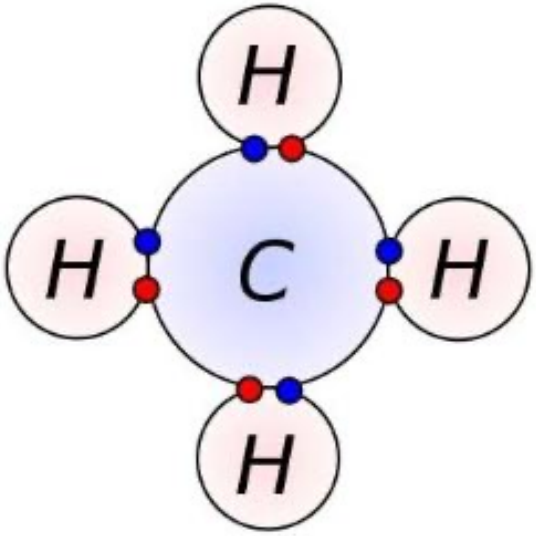
Chemistry of life : Chemical bonding

- Three Main Types of Chemical Bonds:

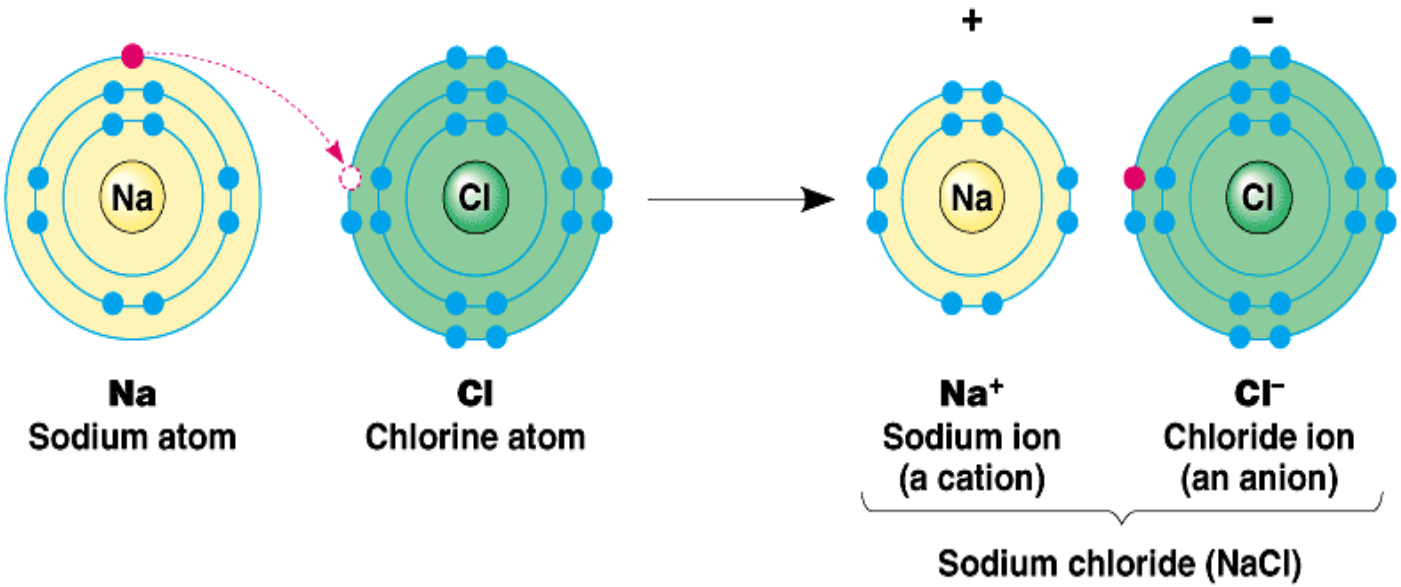
  1. Ionic
  2. Covalent
  3. Hydrogen







Chemistry of life : Ionic Bonding



- Ionic bond involves transfer of electrons between two atoms. Found mainly in inorganic compounds.

- Ion = an atom or group of atoms which have lost or gained one or more electrons, making them negatively or positively charged.

Q: What are positively charged ions (+) called?

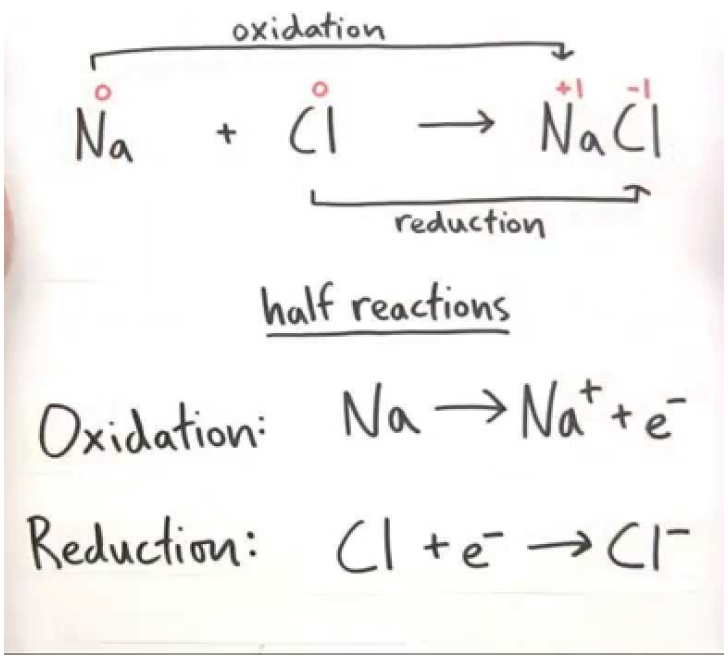
Q: What are negatively charged ions (-) called?

Chemistry of life : Oxidation and Reduction

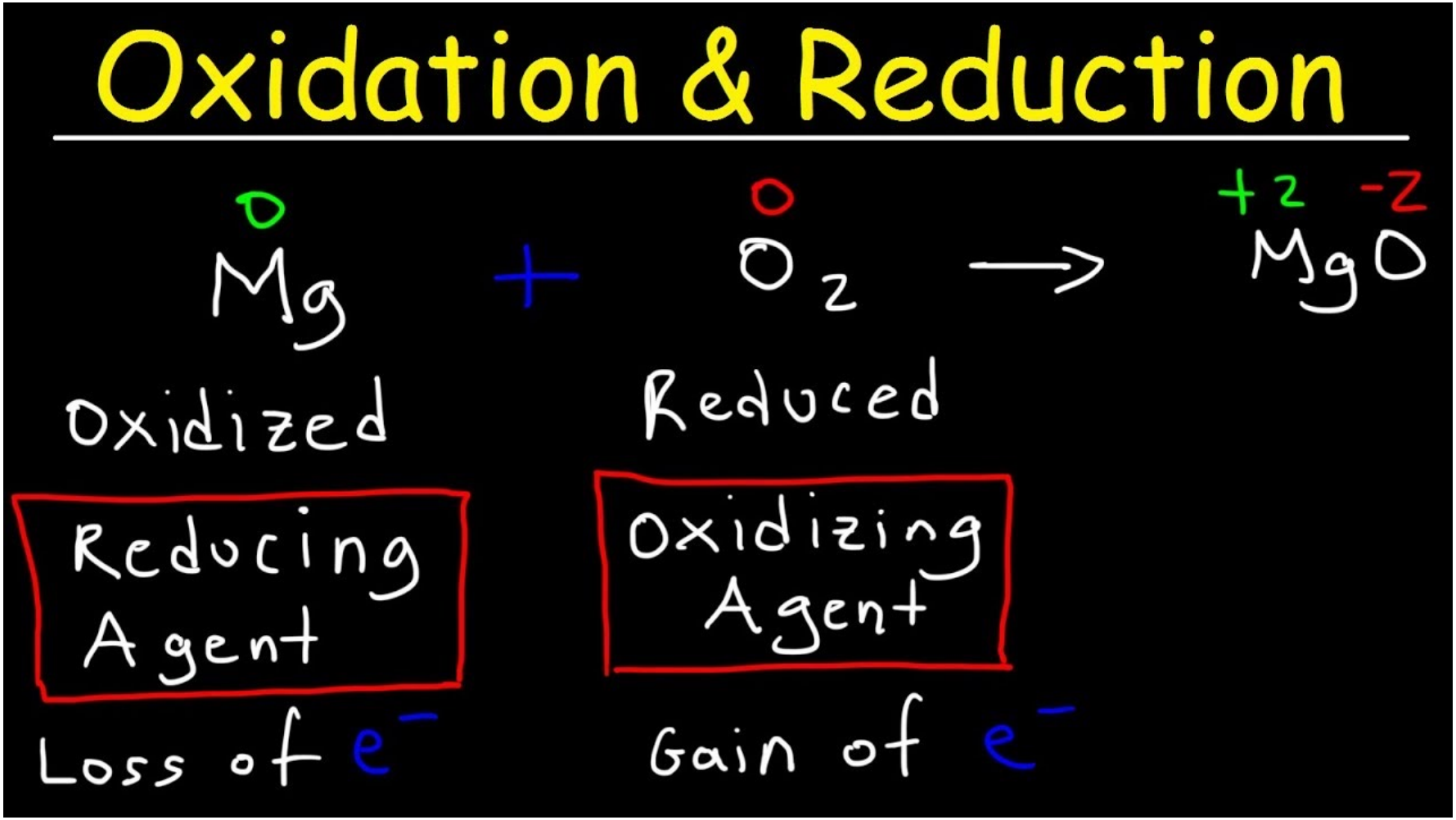
- Oxidation: loss of electrons by a molecule, atom or ion.

- Reduction: gain of electrons by a molecule, atom or ion.

- Redox reaction = chemical reactions in which electrons are gained, lost in a chemical reaction.



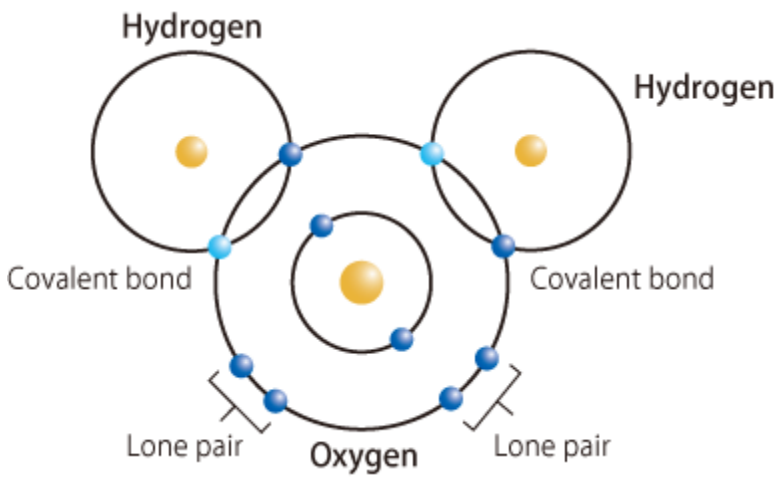
Chemistry of life : Oxidation and Reduction

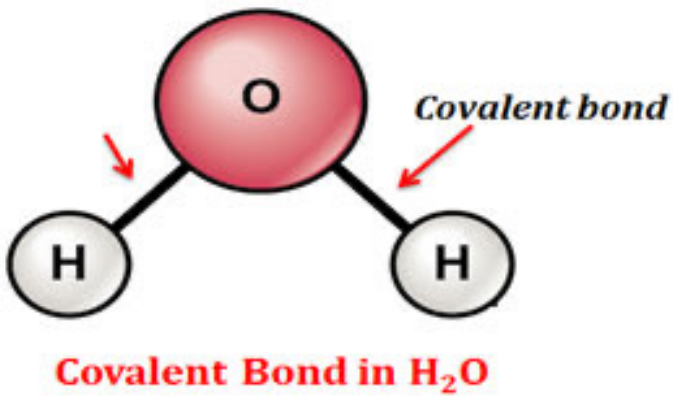


Chemistry of life : Covalent Bonding

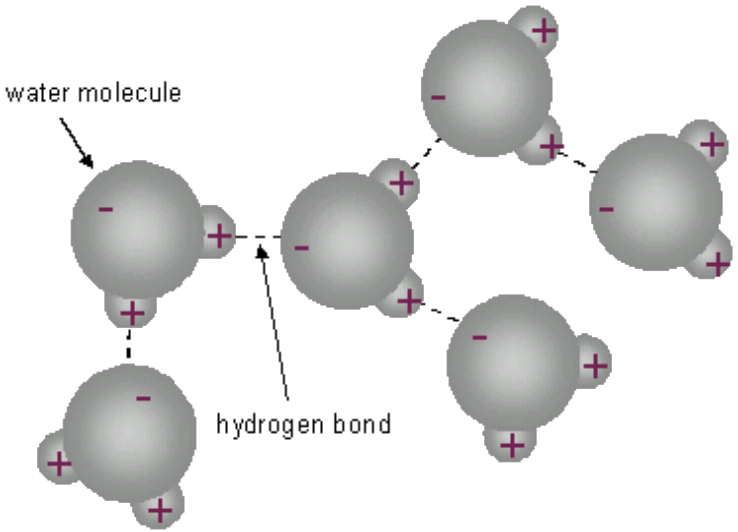
- Covalent bonds involve the sharing of a pair of electrons between two atoms.

- Found mainly in organic compounds.





Chemistry of life : Hydrogen Bonding

○ Hydrogen Bonds: When an atom of hydrogen is attracted to another electronegative atom in addition to the one it is covalently bonded to.

○ In some covalent bonds electrons are **shared unequally** by the hydrogen and the atom that the hydrogen is bound to. When the electrons in a covalent bond are not equally shared, **the molecule is polar.**